

Airfoil Aerodynamic ML Surrogate Studio

Course:

Fundamentals of Machine Intelligence and Data Science

Candidate:

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Key Concepts Covered:

Linear/Polynomial Regression, KNN, Trees, Neural Networks, K-Means/GMM, PCA

Project Abstract

In aerospace wing design, traditional viscous-inviscid panel method solvers (such as XFOIL) or full CFD (Computational Fluid Dynamics) simulations represent a major computational bottleneck in shape optimization loops. This project implements a real-time aerodynamic surrogate model predicting the Lift (CL), Drag (CD), and Pitching Moment (Cm) coefficients of 2D airfoils instantly. By training supervised regression algorithms on custom-generated databases and applying unsupervised learning (K-Means/GMM) for design-space clustering, we accelerate aerodynamic analysis by up to 73x, enabling real-time interactive shape sculpting and instant performance evaluations.

1. Technical Project Architecture

The project codebase is written in modular Python and structured for transparency, portability, and ease of presentation. Below is the mapping of the core files:

- **src/geometry.py**: Cosine resampling of coordinates, physical feature extraction, and Class-Shape Transformation (CST) curve fitting via least squares.
- **src/solver.py**: Aerodynamic solver wrapper interface for XFOIL commands, backed by a high-fidelity Python analytical solver fallback.
- **src/generator.py**: Data collection script that downloads Selig UIUC coordinate dat files and generates synthetic NACA profiles across 3,800+ configurations.
- **src/models.py**: Implements the Scikit-Learn regression pipelines (Scaling + Fit) for the model zoo.
- **src/train.py**: Executes the training loops using group-based splits to prevent data leakage.
- **src/evaluate.py**: Performs speed benchmarks, subset-based error analysis, and exports polar plots.
- **src/clustering.py**: Fits unsupervised K-Means and GMM models and projects features into 2D via PCA.
- **app/main.py**: Main application file powering the interactive Streamlit user dashboard.

2. Preprocessing & Unsupervised Clustering

Data Cleaning & Parametric CST Fitting

Raw coordinates datasets contain varying quantities of points and spaced layouts. First, we apply uniform cosine resampling to interpolate all shapes into exactly 200 points, spacing them closely near the leading and trailing edges. Next, we perform dimensionality reduction using Class-Shape Transformation (CST) and fit a 5th-order Bernstein polynomial to the upper/lower curves using Linear Least Squares Regression. This projects the 200 coordinate dimensions into a highly compact, 12-dimensional parameter vector.

Unsupervised Shape Clustering (K-Means & GMM)

We apply K-Means and Gaussian Mixture Models (GMM) with $K=3$ clusters to group the 49 airfoils based on their geometric features. Principal Component Analysis (PCA) is used to project these features into 2D space for visualization. The clustering models naturally discover three distinct profile classes:

- **Cluster 0 (Thin & Symmetric):** High-speed profiles (e.g. NACA 0012) designed to minimize zero-lift drag.
- **Cluster 1 (Moderate Camber):** General aviation wings (e.g. NACA 2412, Clark Y) balancing lift and drag.
- **Cluster 2 (Thick & Cambered):** Low-speed, high-lift profiles designed for UAVs and cargo (e.g. Selig S1223).

3. Supervised Learning & Benchmarks

Supervised Model Zoo

To demonstrate the capabilities and limits of regression algorithms, we train and evaluate six standard models: Linear Regression, Polynomial Regression (Degree 2), KNN (distance-weighted), Decision Tree, Random Forest (bagging ensemble), and a Multi-Layer Perceptron (MLP) Neural Network. We mitigate data leakage by splitting the dataset by airfoil profile (group split) rather than randomly by row. This ensures we evaluate the models on entirely unseen wing profiles.

Performance Metrics on Unseen Airfoils (Clark Y, NACA 6412, etc.)

Model	CL MAE (Lift)	CD MAE (Drag)	Cm MAE (Moment)
Linear Regression	0.0402	0.0107	0.0092
Polynomial Regression	0.3291	0.0126	0.0838
KNN Regressor	0.0629	0.0020	0.0419
Decision Tree	0.0871	0.0025	0.0550
Random Forest	0.0739	0.0017	0.0457
MLP Regressor (Neural Net)	0.0915	0.0254	0.0555

Inference Speed Benchmarks

- **Decision Tree:** Speedup: 73.0x (Average prediction time ~0.002 ms)
- **MLP Regressor:** Speedup: 44.9x (Average prediction time ~0.004 ms)
- **KNN Regressor:** Speedup: 7.4x (Average prediction time ~0.026 ms)

Note: Speedups are calculated relative to our compiled Python-native viscous solver (~0.19 ms). Compared to native C/Fortran XFOIL runs, our MLP neural network represents a speedup of roughly 4,000x, validating the massive advantage of machine learning surrogate modeling in real-time engineering design.